

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

20th March 2018

Subject CT2 – Finance and Financial Reporting

Time allowed: Three Hours (10.30 – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. Mark allocations are shown in brackets.*
- 3. Attempt all questions, beginning your answer to each question on a separate sheet. However, answers to objective type questions could be written on the same sheet.*
- 4. Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** Cost of equity is
- A. The risk free rate plus equity risk premium
 - B. The opportunity cost of capital
 - C. One of the determinants of weighted average cost of capital
 - D. All of the above
- [2]
- Q. 2)** What is the primary goal of the financial manager of an organization?
- A. To avoid volatility in returns
 - B. To increase the market value of each shareholder's stake
 - C. To comply with various internal and regulatory requirements
 - D. To maximize dividend payouts
- [2]
- Q. 3)** An Australian company entered into a futures contract to exchange Australian dollars for one million US dollars on 31 December 2017. Which of the following will happen if the US dollar strengthens against the Australian dollar?
- A. The Australian company will have to pay an additional margin to the counterparty.
 - B. The Australian company will have to pay an additional margin to the futures clearing house.
 - C. The Australian company will receive a partial refund of its margin payment from the counterparty.
 - D. The Australian company will receive a partial refund of its margin payment from the futures clearing house.
- [2]
- Q. 4)** In India, Capital gains Tax (CGT) is computed after application of Cost Inflation Index (CII) on the purchase price. Calculate the CGT payable at the rate of 20% by an investor who purchased a property in June 2007 and sold it in June 2015, based on the following CII.

Financial year	CII
2006-07	519
2007-08	551
2008-09	582
2009-10	632
2010-11	711
2011-12	758
2012-13	852
2013-14	939
2014-15	1024
2015-16	1081
2016-17	1125

Purchase price = Rs 10,00,000

Sale price = Rs 25,00,000

- A. Nil
B. Rs 91,651.54
C. Rs 1,07,622.50
D. Rs 15,00,000.00 [2]
- Q. 5) Company X based in the UK has an overseas subsidiary, which makes gross profits of Rs 10m. These profits are taxed at 15% tax in the overseas territory. Assuming that the corporation tax rate is 20% in the UK and that a double taxation agreement is in force with the overseas territory, Company X will have to pay:
- A. No further tax on the £8.5m net profit.
B. A further 5% tax on the £10m gross profits.
C. A further 5% tax on the £8.5m net profits.
D. A further 20% tax on the £10m gross profits. [2]
- Q. 6) Which one of the following is the correct formula for the price earnings ratio?
- A. Market price of share/ Earning per share
B. Issue price of share/ Earning per share
C. Market price of share* Earning per share
D. Issue price of share* Earning per share [2]
- Q. 7) While calculating Inventory turnover period, Inventories include
- A. Finished goods
B. Work-in-progress
C. Raw material
D. All of the above [2]
- Q. 8) Which of the following is true?
- A. General insurance companies are more important in medium and long term investment market compared to life insurance companies.
B. General insurance company's liabilities are not fixed in nature for their most lines of business.
C. Life insurance companies are not the important investors in gilts and equities compared to pension funds.
D. All of the above [2]
- Q. 9) Which of the following statement is not correct?
- A. An investment trust is a closed end fund.
B. A unit trust is regulated by trust law.
C. Price of shares in an investment trust are not determined by markets.
D. A unit trust is regulated by trust law. [2]

- Q. 10)** Which option is the correct one?
- A. IRDAI is a banking regulator in India.
 - B. SEBI is an insurance regulatory body in India.
 - C. PFRDA is a pension fund regulator in India.
 - D. All of the above
- [2]
- Q. 11)** State Modigliani and Miller's irrelevance propositions. [2]
- Q. 12)**
- i) Explain the possible reasons for Indian insurance companies to be listing on the stock exchange. (5)
 - ii) What are the disadvantages of listing? (2)
 - iii) Name two Indian life insurance companies that have listed on the stock exchange recently. Which method was adopted? (2)
- [9]
- Q. 13)** Explain the likely impact on company share price in the following scenarios:
- i) Imposition of a long-term capital gains tax on gains from equity (1)
 - ii) Declaration of an interim dividend (1)
 - iii) Company raising additional capital via loan financing (debt) (2)
- [4]
- Q. 14)** A listed company is planning to make a rights issue on the basis of one new share for every five shares currently held. The present share price is Rs 218. The rights issue will be priced at Rs 185. The directors intend to use the funds raised to fund a project that is expected to increase the company's present market capitalisation by 20%. Calculate the expected price per share after the rights issue. Is the rights issue expected to benefit the investors? [5]
- Q. 15)**
- i) State the formula for the beta of a stock. (1)
 - ii) State possible reasons why stocks trade at different betas in the stock market. (1)
 - iii) A company has a geared beta of 1.1 on a debt equity ratio of 1:2 and corporate tax rate of 30%. Calculate the geared beta if the debt equity ratio is 2:2. (3)
- [5]
- Q. 16)**
- i) An individual investor has the following three investment options. The risk free rate prevailing in the market is 7%.

Investment 1: Invest Rs 10,00,000 at the start of year 1, Rs 18,00,000 payout at the start of year 10

Investment 2: Invest Rs 2,00,000 at the start of each of the years 1,2,3,4,5. Payouts of Rs 3,00,000 at the start of years 6,7,8,9,10.

Investment 3: Invest Rs 10,00,000 at the start of year 1. Payouts of Rs 5,00,000 at the start of years 4,7,10.

Evaluate the three investment options using an NPV approach. (5)

ii) Option 4 is an investment option where returns are not fully guaranteed but linked to a market index. The investment structure is as under Option 2 for the first 5 years. In the best estimate scenario, at the start of year 6, Rs 2,70,000 is payable which will increase at 8% p.a. thereafter until the start of year 10. The actual payouts, however, will be linked to the index movement at the time of payouts.

a) Comment on the outcome from this model, assuming a risk margin of 1% for the possible variability in payout. (3)

b) Rather than adjusting the Risk Discount Rate, what could be the other approach to evaluate this index-linked investment option? (2)

iii) What could be some of the other considerations in choosing the investment option other than the NPV approach? (3)

[13]

Q. 17) Indian life insurance companies are entering into derivative contracts to receive fixed rate payments against a floating rate.

i) What would be the key reason behind this? (1)

ii) What other factors could influence the decision to use derivatives? (1)

[2]

Q. 18) **i)** Briefly describe why it is more complicated to prepare insurance companies accounts as compared to normal company accounts. (1)

ii) List and define the major component of reserves for a general insurance company's technical account. (4)

[5]

Q. 19) The details of the share capital and reserves of Company ABC and XYZ is provided in the table below. The par value of shares in Company ABC and XYZ are INR 80 and INR 20.

Figures in INR	Company ABC	Company XYZ
Share Capital	5,00,000	2,00,000
Reserves	5,00,000	50,000

Calculate the goodwill cost of control in two different scenarios if Company ABC shares are priced at par and Company XYZ shareholders are offered 1 share in ABC for every 2 shares in XYZ.

i) 100% of Company XYZ (2)

ii) 50% of Company XYZ (2)

[4]

- Q. 20)** i) Briefly explain the realization and accruals concepts of accounting concepts. (1)
- ii) A Health insurance company sells one year medical plans throughout the year. Describe the specific ways the realization and accrual concepts is utilized in preparing the accounts of the company. (3)
- Q. 21)** Explain depreciation and the methods for depreciation adjustment. [4]
- Q. 22)** Briefly list down the limitation of ratio analysis [4]
- Q. 23)** You have provided following information for a Health Insurance Company called "Moon Healthcare". You are requested to prepare the revenue account and profit and loss account for the year ended 31st March 2017 and the balance sheet as on that date.

The following information are provided for the year, start and end of FY 2017 (figures are in INR millions) –

1. Health fund carried forward (as on 1st April 2016) is 1,000.
2. General reserves amount at start of FY 2017 is 50.
3. Investments are of value 4,615 and income from investment is 150 during the year.
4. Premium income from policyholders from direct business for the year is 4,000 and claims for the amount 600 has been paid during the year. The outstanding premium from direct business is 20.
5. Company has a reinsurance program where Moon Healthcare pays out 100 to reinsurance as reinsurance premium and claims recovered from reinsurance is 20.
6. Company received 50 as commission on reinsurance ceded portion and pays 10 to broker for placing reinsurance. The commission on direct business is 300.
7. The expense of management for Moon Healthcare is 300.
8. The outstanding claim reserves including IBNR as on 31st March 2017 is 100.
9. Transfer of 200 to be made from current profit to General reserves.
10. The company maintains all the reserves and there is an additional reserves for unexpired risk @50% of net premium income (i.e. net of re-insurance).
11. Tax to be provided at 30%.

There are few other information provided (figures are in INR millions) –

Share Capital	900
P&L Account (b/d) brought forward	50
Audit fees	40
Rents	60
Sundry creditor	25
Cash in hand and Bank balances	200

[20]
